Accelerated Algebra and Analysis: Algebra Part

Angus

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Abstract

This is a simple abstract for the Accelerated Algebra and Analysis notes it is focused on the algebra part.

1 Lecture 1

Vectors, Matrices, and Complex numbers

1.1 Geometric motivation

Cubic curves are plain curves given by

$$f(x, y) = 0 \quad \text{where } f \text{ is a cubic polynomial in } x \text{ and } y$$

Theorem 1.1

Chasles's theorem

Let C_3 and C_3' be two cubic curves such that their intersection $C_3 \cap C_3'$ consists of 9 points. Let Z_3 be the set of 8 intersection points

$$Z_3 = (C_3 \cap C_3') \setminus \{P\}$$

where P is the ninth intersection point. Then any cubic curve passing through the points of Z_3 also passes through P.

Chasles's theorem implies Pascal's theorem.

Theorem 1.2

Pascal's theorem

Let A, B, C, D, E, F be the six vertices of a hexagon inscribed in a conic. Then the points of intersection of the pairs of opposite sides $(AB \cap DE), (BC \cap EF), (CD \cap FA)$ are collinear.

- Hexagrammum Mysticum.
- Pappus line

1.2 Vectors

1.2.1 Multidimensional vectors

An n-dimensional vector is an ordered list with n components. It can be represented as

$$\mathbf{v} = (v_1, v_2, \ldots, v_n)$$

We usually write this vector as a column vector

$$\mathbf{v} = \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{pmatrix}$$

Notation

In handwritten notes, an n-dimensional vector $\mathbf{x}$ is often denoted by

$$\overline{x} \quad \underline{x} \quad \mathbf{x} \quad \vec{x}$$

Notation

The set of all n-dimensional vectors is denoted by $\mathbb{R}^n$

$$\mathbb{R}^n = \{(v_1, v_2, \ldots, v_n) \mid v_i \in \mathbb{R} \text{ for } i = 1, 2, \ldots, n\}$$

A special vector in $\mathbb{R}^n$ is the zero vector

$$\mathbf{0} = (0, 0, \ldots, 0)$$

1.2.2 Operations with vectors

Let $\mathbf{x} = (x_1, x_2, \ldots, x_n)$ and $\mathbf{y} = (y_1, y_2, \ldots, y_n)$ be two vectors in $\mathbb{R}^n$.

- **Vector addition:**

$$\mathbf{x} + \mathbf{y} = \begin{pmatrix} x_1 + y_1 \\ x_2 + y_2 \\ \vdots \\ x_n + y_n \end{pmatrix}$$

- **Scalar multiplication:**

$$c\mathbf{x} = \begin{pmatrix} cx_1 \\ cx_2 \\ \vdots \\ cx_n \end{pmatrix} \quad \text{for } c \in \mathbb{R}$$

Let $\mathbf{x}, \mathbf{y}, \mathbf{z} \in \mathbb{R}^n$. The basic properties of vector operations are

1. $\mathbf{x} + \mathbf{y} = \mathbf{y} + \mathbf{x}$
2. $(\mathbf{x} + \mathbf{y}) + \mathbf{z} = \mathbf{x} + (\mathbf{y} + \mathbf{z})$

3. $c(\mathbf{x}+\mathbf{y}) = c\mathbf{x} + c\mathbf{y}$ for $c \in \mathbb{R}$
4. $(cd)\mathbf{x} = c(d\mathbf{x})$ for $c, d \in \mathbb{R}$
5. $\mathbf{x} + \mathbf{0} = \mathbf{x}$
6. $\mathbf{x} + (-\mathbf{x}) = \mathbf{0}$
7. $1\mathbf{x} = \mathbf{x}$ for all $\mathbf{x} \in \mathbb{R}^n$
8. $0\mathbf{x} = \mathbf{0}$ for all $\mathbf{x} \in \mathbb{R}^n$

1.2.3 Length and distance

Definition 1.1

The length (or norm) of a vector $\mathbf{x} \in \mathbb{R}^n$ is defined as

$$\|\mathbf{x}\| = \sqrt{x_1^2 + x_2^2 + \cdots + x_n^2}$$

Definition 1.2

The distance between two vectors $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ is defined as

$$d(\mathbf{x}, \mathbf{y}) = \|\mathbf{x} - \mathbf{y}\|$$

Let $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$. Then the distance between $\mathbf{x}$ and $\mathbf{y}$ satisfies the following properties

1. $d(\mathbf{x}, \mathbf{y}) \geq 0$ and $d(\mathbf{x}, \mathbf{y}) = 0$ if and only if $\mathbf{x} = \mathbf{y}$
2. $d(\mathbf{x}, \mathbf{y}) = d(\mathbf{y}, \mathbf{x})$
3. $d(\mathbf{x}, \mathbf{z}) \leq d(\mathbf{x}, \mathbf{y}) + d(\mathbf{y}, \mathbf{z})$ (triangle inequality)
4. $d(a\mathbf{x}, a\mathbf{y}) = |a| d(\mathbf{x}, \mathbf{y})$ for $a \in \mathbb{R}$

1.2.4 Dot product

Definition 1.3

Let $\mathbf{x}, \mathbf{y} \in \mathbb{R}^3$. The dot product of $\mathbf{x}$ and $\mathbf{y}$ is defined by

$$\mathbf{x} \cdot \mathbf{y} = x_1 y_1 + x_2 y_2 + x_3 y_3$$

Let $\mathbf{x}, \mathbf{y}, \mathbf{z} \in \mathbb{R}^n$ and $c \in \mathbb{R}$. The dot product satisfies the following properties

1. $\mathbf{x} \cdot \mathbf{y} = \mathbf{y} \cdot \mathbf{x}$ (commutativity)
2. $(\mathbf{x} + \mathbf{y}) \cdot \mathbf{z} = \mathbf{x} \cdot \mathbf{z} + \mathbf{y} \cdot \mathbf{z}$ (distributivity)
3. $(c\mathbf{x}) \cdot \mathbf{y} = c(\mathbf{x} \cdot \mathbf{y})$ (scalar multiplication)
4. $\mathbf{x} \cdot \mathbf{x} = \|\mathbf{x}\|^2$
5. $\mathbf{x} \cdot \mathbf{0} = 0$

- If $\mathbf{x} \cdot \mathbf{y} = 0$, we call $\mathbf{x}$ and $\mathbf{y}$ orthogonal
- If $|\mathbf{x} \cdot \mathbf{y}| = \|\mathbf{x}\|\|\mathbf{y}\|$, they are collinear

Since

$$\mathbf{x} \cdot \mathbf{y} = \|\mathbf{x}\|\|\mathbf{y}\| \cos\theta$$

1.2.5 Cauchy-Schwarz inequality

Theorem 1.3

Cauchy-Schwarz inequality

Let $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$. The Cauchy-Schwarz inequality states that

$$|\mathbf{x} \cdot \mathbf{y}| \leq \|\mathbf{x}\|\|\mathbf{y}\|$$

with equality if and only if $\mathbf{x}$ and $\mathbf{y}$ are linearly dependent.

Proof. Consider the function

$$f(\lambda) = \|\mathbf{x} + \lambda\mathbf{y}\|^2$$

Since the norm is always non-negative, we have $f(\lambda) \geq 0$ for all $\lambda \in \mathbb{R}$.

$$\|\mathbf{x} + \lambda\mathbf{y}\|^2 = (\mathbf{x} + \lambda\mathbf{y}) \cdot (\mathbf{x} + \lambda\mathbf{y}) = \mathbf{x} \cdot \mathbf{x} + 2\lambda(\mathbf{x} \cdot \mathbf{y}) + \lambda^2(\mathbf{y} \cdot \mathbf{y}) \geq 0$$

This is a quadratic inequality in λ.
For it to hold for all $\lambda \in \mathbb{R}$, the discriminant must be non-positive:

$$(2(\mathbf{x} \cdot \mathbf{y}))^2 - 4(\mathbf{y} \cdot \mathbf{y})(\mathbf{x} \cdot \mathbf{x}) \leq 0$$

Simplifying, we get

$$(\mathbf{x} \cdot \mathbf{y})^2 \leq (\mathbf{x} \cdot \mathbf{x})(\mathbf{y} \cdot \mathbf{y}) = \|\mathbf{x}\|^2\|\mathbf{y}\|^2$$

which is the Cauchy-Schwarz inequality. □

1.2.6 Triangular inequality

Theorem 1.4

Triangular inequality

Let $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$. The triangular inequality states that

$$\|\mathbf{x} + \mathbf{y}\| \leq \|\mathbf{x}\| + \|\mathbf{y}\|$$

with equality if and only if one vector is a non-negative scalar multiple of the other.

Proof. Consider the square of the norm of the sum:

$$\|\mathbf{x} + \mathbf{y}\|^2 = (\mathbf{x} + \mathbf{y}) \cdot (\mathbf{x} + \mathbf{y}) = \|\mathbf{x}\|^2 + 2(\mathbf{x} \cdot \mathbf{y}) + \|\mathbf{y}\|^2$$

Using the Cauchy-Schwarz inequality, we have

$$2(\mathbf{x} \cdot \mathbf{y}) \leq 2\|\mathbf{x}\|\|\mathbf{y}\|$$

Therefore

$$\|\mathbf{x}+\mathbf{y}\|^2 \leq \|\mathbf{x}\|^2 + 2\|\mathbf{x}\|\|\mathbf{y}\| + \|\mathbf{y}\|^2 = (\|\mathbf{x}\| + \|\mathbf{y}\|)^2$$

Taking the square root of both sides gives the triangular inequality

$$\|\mathbf{x}+\mathbf{y}\| \leq \|\mathbf{x}\| + \|\mathbf{y}\|$$

□

1.2.7 Angles between vectors

Definition 1.4

The angle θ between two non-zero vectors $\mathbf{x}$ and $\mathbf{y}$ is defined by the formula

$$\cos\theta = \frac{\mathbf{x}\cdot\mathbf{y}}{\|\mathbf{x}\|\|\mathbf{y}\|}$$

Lemma 1.5

Pythagoras theorem for vectors states that if two vectors $\mathbf{x}$ and $\mathbf{y}$ are orthogonal, then

$$\|\mathbf{x}+\mathbf{y}\|^2 = \|\mathbf{x}\|^2 + \|\mathbf{y}\|^2$$

If $\mathbf{x}$ and $\mathbf{y}$ are orthogonal (perpendicular), then

$$\mathbf{x}\cdot\mathbf{y} = 0$$

and

$$\theta = \frac{\pi}{2}$$

1.2.8 Cross product of three-dimensional vectors

Definition 1.5

Let $\mathbf{x} = (x_1, x_2, x_3)$ and $\mathbf{y} = (y_1, y_2, y_3)$ be vectors in $\mathbb{R}^3$.
The cross product $\mathbf{x}\times\mathbf{y}$ is defined as

$$\mathbf{x}\times\mathbf{y} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ x_1 & x_2 & x_3 \\ y_1 & y_2 & y_3 \end{vmatrix}$$

where $\mathbf{i}, \mathbf{j}, \mathbf{k}$ are the standard unit vectors in $\mathbb{R}^3$.

$$\mathbf{x}\times\mathbf{y} = \begin{pmatrix} x_2y_3 - x_3y_2 \\ -x_1y_3 + x_3y_1 \\ x_1y_2 - x_2y_1 \end{pmatrix}$$

The cross product has the following properties:

1. $\mathbf{x}\times\mathbf{y} = -(\mathbf{y}\times\mathbf{x})$
2. $\mathbf{x}\times(\mathbf{y}+\mathbf{z}) = \mathbf{x}\times\mathbf{y} + \mathbf{x}\times\mathbf{z}$
3. $(a\mathbf{x})\times\mathbf{y} = a(\mathbf{x}\times\mathbf{y})$ for $a \in \mathbb{R}$

4. $\|\mathbf{x} \times \mathbf{x}\| = 0$
5. $(\mathbf{x} \times \mathbf{y}) \cdot \mathbf{x} = 0$ and $(\mathbf{x} \times \mathbf{y}) \cdot \mathbf{y} = 0$
6. $\|\mathbf{x} \times \mathbf{y}\| = \|\mathbf{x}\|\|\mathbf{y}\| \sin\theta$, where θ is the angle between $\mathbf{x}$ and $\mathbf{y}$

1.3 Matrix

1.3.1 Definition

Definition 1.6

A matrix is a rectangular array of numbers arranged in rows and columns. An $m \times n$ matrix has m rows and n columns and is typically denoted by

$$A = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{pmatrix}$$

Zero matrix is a matrix in which all the elements are zero, typically denoted by

$$0 = \begin{pmatrix} 0 & 0 & \dots & 0 \\ 0 & 0 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 0 \end{pmatrix}$$

$n \times n$ identity matrix is a matrix in which all the elements on the main diagonal are 1 and all other elements are 0, typically denoted by

$$I_n = \begin{pmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{pmatrix}$$

1.3.2 Sum representation

Any $m \times n$ matrix $A = (a_{ij})$ can be represented as the sum of its elements multiplied by the corresponding standard basis matrices

$$A = \sum_{i=1}^{m} \sum_{j=1}^{n} a_{ij} E_{ij},$$

where E_{ij} is the $m \times n$ matrix with 1 in the (i, j)-th position and 0 elsewhere.

1.3.3 Algebra operations with matrices

$A + B$

$$(A + B)_{ij} = A_{ij} + B_{ij}$$

cA

$$(cA)_{ij} = c \cdot A_{ij}$$

AB

$$(AB)_{ij} = \sum_{k=1}^{n} A_{ik} B_{kj}$$

1.3.4 Matrix multiplication

Definition 1.7

matrix multiplication

$$A = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{pmatrix} \quad B = \begin{pmatrix} b_{11} & b_{12} & \dots & b_{1q} \\ b_{21} & b_{22} & \dots & b_{2q} \\ \vdots & \vdots & \ddots & \vdots \\ b_{p1} & b_{p2} & \dots & b_{pq} \end{pmatrix}$$

$$AB = \begin{pmatrix} (AB)_{11} & (AB)_{12} & \dots & (AB)_{1q} \\ (AB)_{21} & (AB)_{22} & \dots & (AB)_{2q} \\ \vdots & \vdots & \ddots & \vdots \\ (AB)_{m1} & (AB)_{m2} & \dots & (AB)_{mq} \end{pmatrix}$$

Make sure that the number of columns of A matches the number of rows of B for the matrix multiplication AB to be defined,

Example

If A is an $m \times n$ matrix and B is a $p \times q$ matrix, then $n = p$. In particular, the product AB will be an $m \times q$ matrix.

$$A = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{pmatrix}_{m \times n}, \quad B = \begin{pmatrix} b_{11} & b_{12} & \dots & b_{1q} \\ b_{21} & b_{22} & \dots & b_{2q} \\ \vdots & \vdots & \ddots & \vdots \\ b_{p1} & b_{p2} & \dots & b_{pq} \end{pmatrix}_{p \times q}$$

$$AB = \begin{pmatrix} (AB)_{11} & (AB)_{12} & \dots & (AB)_{1q} \\ (AB)_{21} & (AB)_{22} & \dots & (AB)_{2q} \\ \vdots & \vdots & \ddots & \vdots \\ (AB)_{m1} & (AB)_{m2} & \dots & (AB)_{mq} \end{pmatrix}_{m \times q}$$

Basic properties of matrix addition and scalar multiplication

1. $(A + B) = (B + A)$
2. $(A + B) + C = A + (B + C)$
3. $A + 0 = A$
4. $c(A + B) = cA + cB, \quad c \in \mathbb{R}$

Basic properties of matrix multiplication

1. $(AB)C = A(BC)$

2. $A(B+C) = AB + AC$
3. $(A+B)C = AC + BC$
4. $AI_n = A = I_m A_{m\times n}$

For $(AB)C = A(BC)$

Proof. Let A be an $m \times n$ matrix, B be an $n \times p$ matrix, and C be a $p \times q$ matrix. Then

$$\begin{aligned}((AB)C)_{ij} &= \sum_{k=1}^{p}(AB)_{ik}C_{kj} \\ &= \sum_{k=1}^{p}\left(\sum_{l=1}^{n}A_{il}B_{lk}\right)C_{kj} \\ &= \sum_{l=1}^{n}A_{il}\left(\sum_{k=1}^{p}B_{lk}C_{kj}\right) \\ &= (A(BC))_{ij}, \quad 1 \le i \le m, \quad 1 \le j \le q\end{aligned}$$

Therefore, $(AB)C = A(BC)$. □

1.3.5 Column Vectors

Definition 1.8

Column vectors $(x_1, x_2, \ldots, x_m)^T$

A column vector is an $m \times 1$ matrix, i.e.,

$$\mathbf{v} = \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_m \end{pmatrix}_{m\times 1}$$

If

$$A = \begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_m \end{pmatrix}$$

Then

$$MA = \begin{pmatrix} (MA)_1 \\ (MA)_2 \\ \vdots \\ (MA)_m \end{pmatrix}$$

1.3.6 Invertible matrices

Definition 1.9

An $n \times n$ matrix A is said to be invertible if there exists an $n \times n$ matrix B such that

$$AB = BA = I_n.$$

The matrix B is called the inverse of A and is denoted by A^{-1}.

A matrix is called singular if it is an $n \times n$ matrix that is not invertible.

Notation

The set of all invertible $n \times n$ matrices is denoted by $GL(n, \mathbb{R})$ if the entries are real numbers.

For 2×2 case, an invertible matrix $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ has an inverse given by

$$A^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix} \quad ad - bc \neq 0.$$

1.3.7 Transpose

Definition 1.10

Let A be an $m \times n$ matrix,

$$A = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{pmatrix}_{m \times n}.$$

The transpose of A, denoted by A^T, is the $n \times m$ matrix obtained by interchanging the rows and columns of A:

$$A^T = \begin{pmatrix} a_{11} & a_{21} & \dots & a_{m1} \\ a_{12} & a_{22} & \dots & a_{m2} \\ \vdots & \vdots & \ddots & \vdots \\ a_{1n} & a_{2n} & \dots & a_{mn} \end{pmatrix}_{n \times m}.$$

Properties of the transpose

1. $(A^T)^T = A$
2. $(A + B)^T = A^T + B^T$
3. $(cA)^T = cA^T$ for any scalar c
4. $(AB)^T = B^T A^T$

Definition 1.11

Symmetric matrix A matrix A is called symmetric if $A^T = A$.

Skew-symmetric matrix A matrix A is called skew-symmetric if $A^T = -A$.

Theorem 1.6

For any two $n \times n$ matrices A and B, the matrix $AB - BA$ is skew-symmetric.

Proof. We need to show that $(AB - BA)^T = -(AB - BA)$.

$$\begin{aligned}(AB - BA)^T &= (AB)^T - (BA)^T \\ &= B^T A^T - A^T B^T \\ &= -(A^T B^T - B^T A^T) \\ &= -(AB - BA).\end{aligned}$$

Hence, $AB - BA$ is skew-symmetric. □

1.4 Complex Number

1.4.1 Definition

We adjoin the imaginary unit i in $\mathbb{R}$, where

$$i^2 = -1$$

We define a complex number as

$$z = a + bi, \quad a, b \in \mathbb{R}.$$

And we denote the set of all complex numbers by $\mathbb{C}$

$$\mathbb{C} = \{a + bi \mid a, b \in \mathbb{R}\}$$

1.4.2 Fundamental theorem of Algebra

Theorem 1.7

Let

$$p(z) = a_n z^n + a_{n-1} z^{n-1} + \cdots + a_1 z + a_0$$

be a non-constant polynomial with complex coefficients.
If $n \geq 1$, then there exists at least one complex number $z_0 \in \mathbb{C}$ such that

$$p(z_0) = 0.$$

In other words, every non-constant polynomial with complex coefficients has at least one root in $\mathbb{C}$.

Lemma 1.8

Let

$$p(z) = a_n z^n + a_{n-1} z^{n-1} + \cdots + a_1 z + a_0$$

be a non-constant polynomial with complex coefficients.
Then $p(z)$ can be factored into linear factors over $\mathbb{C}$, i.e.,

$$p(z) = a_n (z - z_1)(z - z_2) \dots (z - z_n),$$

where $z_1, z_2, \dots, z_n \in \mathbb{C}$ are the roots of $p(z)$.

1.5 Field

1.5.1 Definition

Definition 1.12

A field is a set F equipped with addition and multiplication, such that the following properties hold

1. F is an abelian group under addition, with identity element 0.
2. $F \setminus \{0\}$ is an abelian group under multiplication, with identity element 1.
3. Multiplication is distributive over addition, i.e., for all $a, b, c \in F$,

$$a(b + c) = ab + ac.$$

Definition 1.13

Abelian group

A group G satisfies the commutative property if for all $a, b \in G$

$$a + b = b + a$$

1.5.2 Other finite fields

To construct many finite fields, fix a prime number $p \geq 2$. Let $\mathbb{F}_p$ be the following set consisting of p integers:

$$\{0, 1, \dots, p - 1\}.$$

Equip $\mathbb{F}_p$ with operations $+$ and $*$ as follows:

1. For every two elements a and b in $\mathbb{F}_p$, let $a + b =$ the remainder of the integer $a + b$ when divided by p.
2. For every two elements a and b in $\mathbb{F}_p$, let $a * b =$ the remainder of the integer $a * b$ when divided by p.

Then $\mathbb{F}_p$ equipped with $+$ and $*$ is a field.

Example

Suppose that $p = 1973$ (it is prime). Then

$$\frac{a}{45} = a * 570$$

for every a in $\mathbb{F}_p$, since $45 * 570 = 1$ in $\mathbb{F}_p$.

1.5.3 Solving system of two linear equations

Example

Let us solve the following system of equations in $\mathbb{F}_7$:

$$\begin{cases} 2x + 3y = 5, \\ 3x + 4y = 1. \end{cases}$$

Observe that $2 \times 4 = 1$ in $\mathbb{F}_7$. Then $\frac{1}{2} = 4$ and

$$2x + 3y = 5 \implies x = 4(5 - 3y),$$

which gives $x = 6 + 2y$. Then

$$3(6 + 2y) + 4y = 1,$$

so that $3y = 4$, which gives $y = 5 \times 4 = 6$. Then

$$(x, y) = (6 + 2 \times 6, 6) = (4, 6).$$

1.5.4 A better way to construct a field with 4 elements

Start with the field $\mathbb{F}_2$ consisting of two elements 0 and 1.
Introduce a new element t such that

$$t^2 = t + 1$$

Consider all sums (there are 4 of them) that look like

$$a + bt$$

where a and b are both in $\mathbb{F}_2$. Denote the obtained set by $\mathbb{F}_4$. Then extend $+$ and $*$ from $\mathbb{F}_2$ to elements in $\mathbb{F}_4$ as follows:

$$(a + bt) * (c + dt) = ac + bdt^2 + adt + bct = (ac + bd) + (ad + bc + bd)t$$

and

$$(a + bt) + (c + dt) = (a + c) + (b + d)t$$

for any elements a, b, c, d in the field $\mathbb{F}_2$. Then $\mathbb{F}_4 = \{0, 1, t, 1 + t\}$ is a field.

2 Lecture 2